

LITERACY
IN
MATHS

MATHS ACROSS THE CURRICULUM

Bob Starink

Quality ready-to-use
resources

INSTANT LESSONS
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Bob Starink

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Bob Starink has been teaching Mathematics for over 25 years at various secondary schools in Western Sydney, NSW. He collaborated with his wife Jennifer, an English teacher, to produce this *Literacy in Maths* series. Bob has co-authored other secondary mathematics textbooks, covering the full range of ability levels. He has been involved in leadership groups and has presented at curriculum workshops. Bob hopes that his *Literacy in Maths* books will improve students' conceptual understanding and make mathematics more than a repetition of processes.

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ABN 074 266 023
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Ph: +61 3 9558 4433
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INTRODUCTION

Understanding the language of maths is critical to understanding mathematical concepts. This literacy resource explores the use of mathematical concepts and calculations across the subjects of the school curriculum. Each topic contains a list of words which are featured in a range of literacy activities for students of all ability levels. The activities provide an emphasis on interpreting, analysing, calculating and appreciating how mathematics is integral to all areas of life.

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1 AGRICULTURE

WORD LIST:

- | | | | | |
|----------|--------------|-------------|------------|---------------|
| • energy | • megajoules | • kilogram | • acidity | • plant |
| • weight | • percentage | • tonne | • alkaline | • animal |
| | • water | • calculate | • hectares | • agriculture |

Use the word list above to complete the following exercises.

THE MATHEMATICS OF FEEDING SHEEP

The energy in feed is measured in megajoules of metabolisable energy (MJ of ME) where ME is the energy available in the feed.

A general rule for calculating the energy required per day per sheep is $0.1 \times \text{animal weight} + 1.8$ MJ of ME.

Two factors that must be allowed for are the digestibility of the feed, which is expressed as a percentage of the feed a sheep can convert to energy, and the moisture in the feed. Feed is sold by weight and includes water. The percentage Dry Matter (DM) is the percentage of the feed that can be used by the sheep to produce energy.

1 A 60 kg sheep is fed with 70% digestible, 80% Dry Matter feed (Food 1) containing 15 MJ of energy per kg.

a Calculate the sheep's required MJ of ME, writing out the mathematics involved.

b How many MJ/kg is available from the feed? Again, write out the mathematics involved.

c What weight of this food (in grams) is required by the sheep per day? Again, write out the mathematics involved.

2 Repeat these calculations for a 46 kg sheep, fed with 50% digestible, 60% Dry Matter feed (Food 2) containing 9 MJ of energy per kg.

3 If Food 1 costs \$270 per tonne and Food 2 costs \$120 per tonne, which is the better value? Why?

4 If Food 1 is chosen and there are 1200 sheep averaging 50 kg in weight, how many tonnes of food will be needed to feed the stock for 30 days? Explain your working.

TRIOS - Soils

The pH scale is divided into 14 units where 7 is neutral, less than 7 means becoming more acidic and more than 7 means becoming more alkaline. Generally, plants need a pH from 4.8 to 8 to grow effectively. A pH below 4.8 is considered acidic. In Australian agriculture, soil acidity is a major problem.

Agricultural land use in Australia	Percentage pH < 4.8	Land area (in hectares)
Natural environment	55%	239 200 000
Crops and horticulture	50%	22 550 000
Grazing	60%	18 500 000
Irrigation	30%	2 030 000

Using the information above, circle the correct term to complete each sentence.

- There are **282 280 000 / 282 880 000 / 282 820 000** hectares of land used for agriculture in Australia.
- The most acidic land is used for **crops / irrigation / grazing**.
- $\frac{1}{3}$ / $\frac{3}{10}$ / $\frac{3}{5}$ of irrigation land area is too acidic.
- 35% / 45% / 55%** of natural environment land has a pH greater than 4.8.
- 11 725 000 / 11 257 000 / 11 275 000** ha of land used for crops or horticulture have acidic soils.
- The total number of hectares of agricultural land in Australia with acidic soils is **154 545 000 / 154 555 000 / 154 544 000**.

A to Z - Write the list words in alphabetical order.

2 CERAMICS

WORD LIST:

- texture
- absorption
- ceramics
- symmetry
- clay
- shrinkage
- materials
- cone
- vertical
- temperature
- composition
- pounds
- radial
- axis
- thickness

Use the word list above to complete the following exercises.

CLAYS

Polymer Clay will harden when baked at low temperature. At 130°C for 15 minutes per 6 mm thickness, no shrinkage or texture change occurs.

- 1 For how long will a 15 mm thick object need to bake at 130°C ? _____
- 2 A Fire Clay has 60% silica, 25% aluminium and small amounts of other minerals. What do you think makes up the rest of the total percentage composition of this clay?

- 3 A China Clay has 12% shrinkage. What is the weight shrinkage of an 8 kg China Clay mould?

- 4 A Buff Clay shrinks from a 10 cm standard length to 9.18 cm. What is the percentage shrinkage?

- 5 A White Clay sample weighs 5.2 g originally and weighs 5.7 g when soaked in water for 24 hours. What is the percentage absorption rate?

TIMELINE

The tables below show the approximate date that some form of ceramic was first made for different countries.

Country	Date
India	5 500 BC
Greece	3 000 BC
Japan	10 500 BC
Holland	1 350 BC
America	200 AD

Country	Date
Korea	7 000 BC
Czech Republic	24 000 BC
China	10 000 BC
Turkey	6 500 BC
Egypt	5 000 BC

Complete the timeline below. First extend the line across the page. Let each centimetre = 1000 years. Label the axis and the countries in their approximate date positions. Then write a scale.



THE POTTER'S WHEEL

When using a potter's wheel to make an object, the finished product has radial symmetry along a vertical axis.

1 What does the underlined description mean?
Draw a diagram in the box to help you explain.

2 What is one disadvantage of using a potter's wheel?

WHICH WORD ...

1 contains a sheep?

2 ends phoning up?

3 has a place to skate inside?

4 starts with a rug?

5 is useful in the UK?

6 begins angrily?

7 is mostly a location?

8 starts with writing?

A to Z - Write the list words in alphabetical order.

3 COMMERCE

WORD LIST:

• consumer	• purchase	• payment	• variable	• wage
• financial	• price	• income	• adviser	• salary
• employment	• credit	• gross	• interest	• commission
• brand	• direct	• net	• rate	• deposit
• generic	• debit	• borrow	• annum	• mortgage
• product	• electronic	• savings	• casual	• percentage
• range	• funds	• fixed	• temporary	• comparative
	• transfer	• annual	• piecework	• tax

Use the word list above to complete the following exercises.

PROCEDURE – Read the following information and complete the procedure below.

Very interesting

After paying a \$40 000 deposit from their savings, Jim and Paula Wogram borrow \$280 000 to buy a house. They opt for an 8.4% per annum fixed interest rate for the first 3 years of their loan. The variable rate of interest at the time of their loan is 7.8% p.a. Jim and Paula's net (after tax) annual household income is \$93 000.

- To determine what percentage the deposit is of the full cost of the house, d _____ 40 000 by the total cost _____, which equals _____%.
- To determine what monthly percentage interest rate the Wograms are paying, d _____ 8.4 by _____, which equals _____%
- To determine the minimum amount of interest they have to pay on their loan in the first month, c _____ % of \$ _____, which is _____.
- To determine how much less they would pay in the first month if they had opted for a variable loan, c _____ which equals \$ _____.
- To determine what percentage of the Wograms' monthly net income their minimum first month repayment is, d _____, which is _____% (to 1 decimal place).
- Financial advisers recommend that no more than 28% of a family's total gross annual income should be used to pay a mortgage. If the Wogram's net income is 75% of their gross income, to determine how much the Wograms can afford to pay each month to make this percentage, calculate their net m _____ income by dividing \$ _____ by 12; work out their monthly g _____ income by m _____ this figure by _____ and then calculate _____% of this amount, which equals \$ _____, to the nearest \$.

7 If they decide to pay this amount each month, to determine how much they paid off their loan after the first month, \$ _____ from \$ _____, which is \$ _____.

WHAT'S THE DIFFERENCE? - What is the difference between:

- 1 a wage and a salary? _____
- 2 benefits and pay rates for temporary and casual employment? _____
- 3 paid commission and piecework? _____
- 4 purchasing by credit card and debit card? _____
- 5 direct debit and EFT (Electronic Funds Transfer)? _____
- 6 fixed and variable interest rates? _____

EVERY THIRD LETTER - Complete the following list words.

- | | | |
|-------------------------|---------------------|-----------------------|
| 1 _ _ M _ _ R _ _ I _ _ | 4 _ _ M _ _ R _ _ Y | 7 _ _ R _ _ B _ _ |
| 2 _ _ N _ _ C _ _ L | 5 _ _ E _ _ W _ _ K | 8 _ _ P _ _ Y _ _ N _ |
| 3 _ _ E _ _ R _ _ I _ | 6 _ _ R _ _ A _ _ | 9 _ _ V _ _ E _ |

HALF THE ALPHABET - Change the bold letter to form a different word. You must use a different letter for each one.

- | | |
|-------------------------|--------------------------|
| 1 b rand _____ | 7 consumer _____ |
| 2 r ange _____ | 8 t ax _____ |
| 3 g ross _____ | 9 f ixed _____ |
| 4 b orrow _____ | 10 r ate _____ |
| 5 d eposit _____ | 11 g eneric _____ |
| 6 w age _____ | 12 p rice _____ |

4 DANCE

WORD LIST:

- | | | | |
|---------------|--------------|-------------|------------|
| • ushers | • venue | • repeat | • reverse |
| • combination | • costumes | • motion | • compass |
| • sequence | • music | • direction | • diagonal |
| • dance | • musicality | • movement | • sound |
| • performance | • time | • natural | • light |
| • catering | • steps | | |

Use the word list above to complete the following exercises.

STAGING A PERFORMANCE

A school stages a dance performance at the local hall. There will be 10 acts to the timed sequence shown in the box at the right.

1 What is the total time the 10 dances take? _____

2 There will be a 25 second delay between each dance, except an extended 45 seconds delay prior to Acts 1, 4 and 7 for introductions and a 65 second applause at the end. What is the overall time of the performance?

3 Doors open at 7 pm, half an hour before the first dance. There are refreshments available afterwards for 20 minutes. The doors are closed 20 minutes later. What time are the doors closed? _____

4 The cost to stage the performance is \$ 1000. Complete the following:

Venue costs would include h ____ h ____, s ____ and l ____, c ____ and us ____ . Dancers' costs would include c ____ and m ____ .

5 The venue holds 250 people. If an 80% attendance is expected, how much should each person be charged in order to break even? _____

- | |
|----------------------|
| • Dance 1: 3 m 23 s |
| • Dance 2: 2 m 15 s |
| • Dance 3: 1 m 41 s |
| • Dance 4: 2 m 35 s |
| • Dance 5: 2 m 19 s |
| • Dance 6: 3 m 54 s |
| • Dance 7: 1 m 37 s |
| • Dance 8: 2 m 04 s |
| • Dance 9: 2 m 46 s |
| • Dance 10: 8 m 11 s |

MUSICALITY – Musicality is fitting dance to music.

1 Waltzes are in $\frac{3}{4}$ time, foxtrots are in $\frac{4}{4}$ time, tangos are in $\frac{2}{4}$ time and Viennese waltzes are in $\frac{6}{8}$ time. The top number represents the number of beats in a bar, or measure, and is often the same as the number of dance steps per bar. Tango can be counted as 1, 2, 1, 2, 1, 2 ...

a What is the waltz count? _____ b What is the foxtrot count? _____

2 Most popular music is in $\frac{4}{4}$ time and follows a pattern called AABA, where A and B represent four 8 count beats, with B having a different melody.

a How many beats are in each bar? _____

- b How many bars in a typical AABA format? _____
- c Where do you think breaks in the music may occur? _____

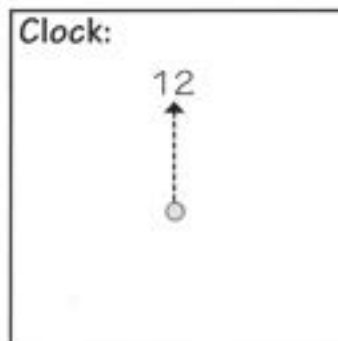
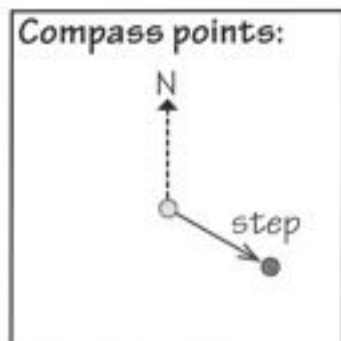
3 Electronic dance music (techno) is usually played between 120 beats per minute and 200 beats per minute. Describe the tempo of this music and the type of dancing that will match it.

4 The steps to the Chicken Dance are shown at the right.

- a How many beats are there per cycle of the dance? _____
- b Considering your answers to Q2, what time do you think this music is played in? _____

DIRECTION OF MOVEMENT

The direction of movement in dance can be written as compass points or like a clock.



- Begin in a circle facing the centre.
- Shape a chicken beak with your hands. Open and close the beak 4 times during the first 4 beats.
- Make chicken wings with your arms. Flap your wings 4 times on the next 4 beats of the song.
- Make a chicken's tail feathers with your arms and hands. Wiggle your tail during the next 4 beats.
- Clap your hands during the next 4 beats.
- Repeat this process a further 3 times.

1 The compass points step would be described as: 'take a step _____'.

2 In the clock box, use an arrow and a new circle to draw where a step of 10 o'clock would go.

3 With respect to the supporting right foot shown at the right, draw where the next step will go with the left foot if the instruction is 'step diagonally forward'.

4 A natural turn is a three-quarter turn to the right. A reverse turn is a one and a quarter turn to the left.

- a Which turn is anticlockwise? _____
- b What is the angle of a natural turn? _____
- c How many degrees are there in a full circle? _____
- d What is the angle of a reverse turn? _____



5 DIGITAL MEDIA

WORD LIST:

- | | | | |
|---------------|-------------|-------------|--------------|
| • pixel | • ratio | • box | • vertical |
| • digital | • sharpness | • aspect | • windowbox |
| • media | • computer | • cropped | • monitor |
| • resolution | • screen | • letterbox | • diagonal |
| • image | • optical | • pillar | • horizontal |
| • composition | | | • telephoto |
| | | | • megapixels |

Use the word list above to complete the following exercises.

SPELL MAD – Underline the incorrectly spelt words in the following passage. Then write the correct spelling above each one.

A zoom lens automaticly changes the focle lenth and the angel of view from wide angl to telefoto. A optical zom maintanes the rezolution of the photo, meening the numbre of pixels is the same and the klarity will bee equal. A digital zoom magnifyfs part off the orignal imige, meaning the skreen is filled with fewer pixles and the sharpnes will detiriorate with increesing zooming. The computer will fill inn the pictur with likly pixals but the overal quality will be less van for an optical zoom set at the saim range.

COMPRENDEZ – Read the passage and answer the questions below. Use the list words to help you.

Aspect ratio

Older style televisions and computer monitors have an aspect ratio of 4:3, meaning for every 4 cm of horizontal length, there is 3 cm of vertical width. Widescreen televisions and computer monitors have an aspect ratio of 16:9. Cinema releases are produced in 1.85:1 and 2.39:1 (both called letterbox). Digital cameras mostly shoot in 4:3. Screens are measured along a diagonal.

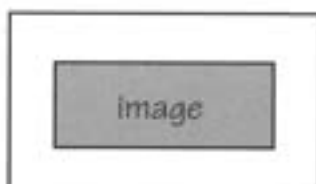
- 1 Express 16:9 as a ratio in the form $x:1$ (to 2 decimal places). _____
- 2 How does a widescreen picture appear as a full screen on a 4:3 television?

- 3 How does a 1.85:1 movie appear on a 16:9 television?

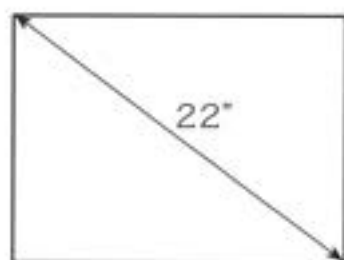
- 4 How does a 4:3 broadcast appear on a 16:9 television?

5 What is this type of image on a widescreen television called?

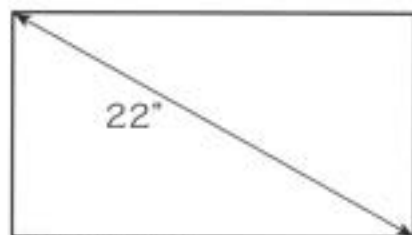
6 What do you think would happen if you printed a 4:3 image onto a 15 by 10 photo paper? Explain your answer mathematically.



7 Two computer monitors are shown below – one is a 22" normal screen, the other is a 22" widescreen. Circle the screen that has the larger area. Then label it to show by how many cm^2 it is larger. Use $1" \approx 2.5 \text{ cm}$, and you will need to use Pythagoras' Theorem and a bit of algebra to work out the answer.



4:3 screen



16:9 screen

IMAGE RESOLUTION

Image resolution is a measure of the detail an image holds. It can be expressed in megapixels, which is the product of the number of horizontal pixels and the number of vertical pixels, divided by a million.

1 Complete the table by calculating the number of megapixels for each device, correct to 2 decimal places.

Device	Horizontal pixels	Vertical pixels	Megapixels
Analogue broadcast	440	480	
Widescreen DVD	720	480	
Blu-Ray	1 920	1 080	
Digital camera	2 048	1 536	
IMAX	10 000	7 000	

A higher megapixel count affects the quality as a picture becomes larger. While a moving picture seems clear, have you tried printing a video capture? You will notice how print quality is affected by a lower megapixel resolution.

6 DRAMA

WORD LIST:	• performance	• design	• arch	• timing
• role	• arena	• theatre-in-	• concentric	• role
• character	• thrust stage	the-round	• act	• venue
• stage	• alter	• proscenium	• play	

Use the word list above to complete the following exercises.

IT'S A NUMBERS GAME – Answer the questions based on the following scenarios.

Dual role:

A class is planning a four-act performance for a school assembly. There are four characters – Peter, Jack, Tom and Steve. In Act 1, Peter, Jack and Steve are on stage. In Act 2, Jack and Tom are on stage. In Act 3, Steve, Tom and Jack are on stage. In Act 4, Tom and Jack are on stage.

1 Sam wishes to play a dual role (2 characters). Which combination of characters is it possible for him to play? _____

One act plays:

A local Drama club is putting on a show with 9 one-act plays. The plays have these lengths:
Act 1 – 3 minutes; Act 2 – 7 minutes; Act 3 – 4 minutes; Act 4 – 7 minutes;
Act 5 – 3 minutes; Act 6 – 8 minutes; Act 7 – 10 minutes; Act 8 – 10 minutes;
Act 9 – 5 minutes.

2 There has to be an intermission two-thirds of the way through the performances. Make one simple rearrangement to the order of the acts so that the correct timing can be achieved.

The play

A major production has these time breakdowns: Act 1 runs for 35 minutes; Acts 2 and 3 run for 20 minutes each, intermission lasts 15 minutes and there is a 5 minute delay before the play starts and after intermission until it continues. The advertised starting time of the play is 7.30 pm.

3 Act 2 begins after intermission. At what time will the actors resume the play? _____

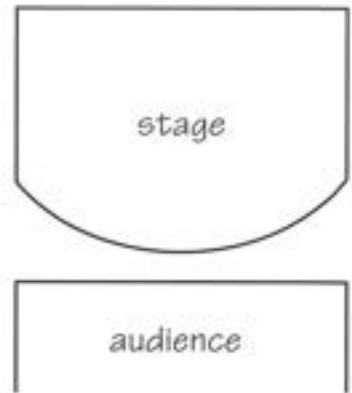
4 What time does the play finish? _____

EVERY THIRD LETTER – Work out the hidden sentence by crossing out every third letter and writing the remaining sentence below.

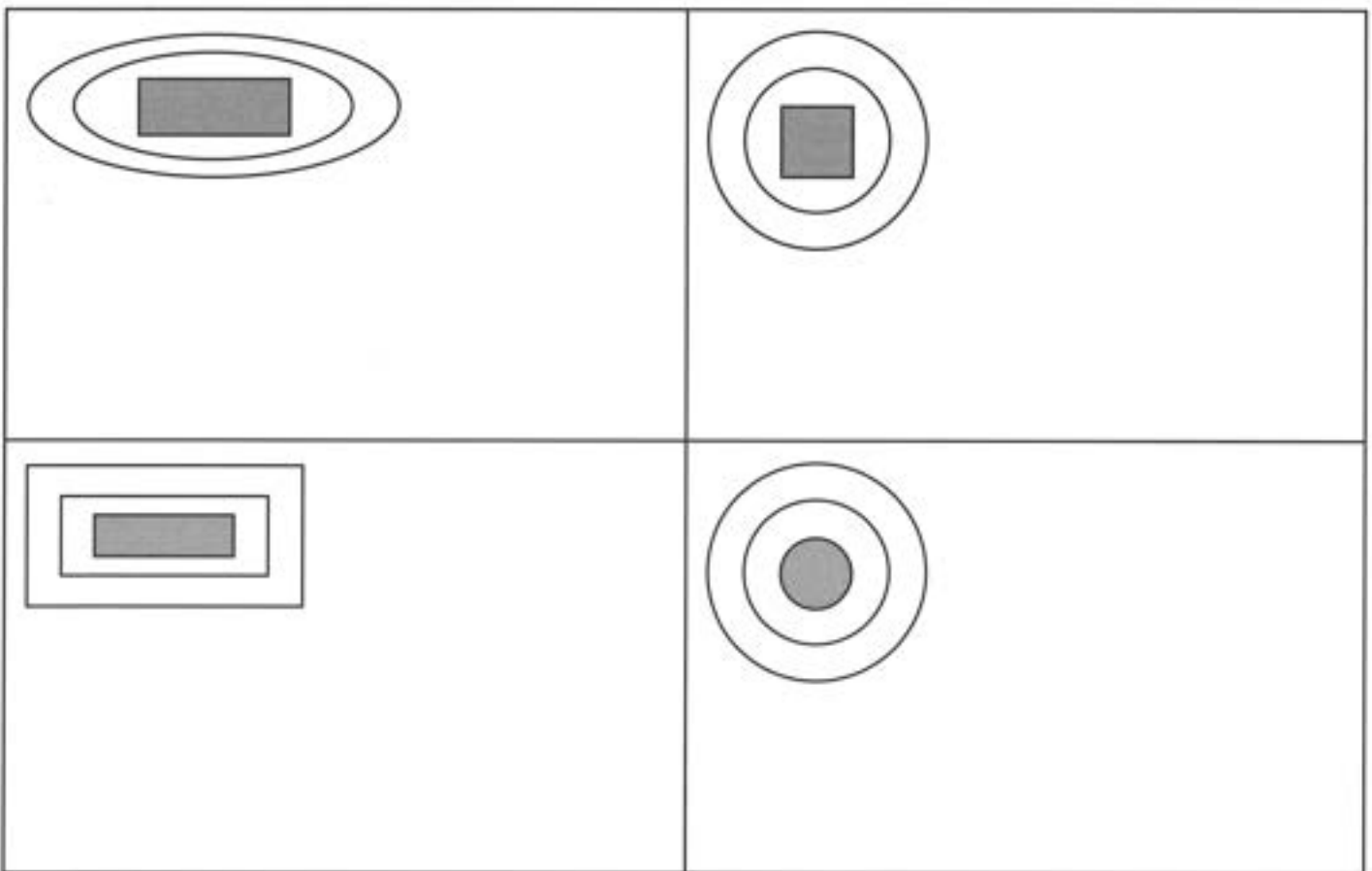
APTROFSCBENAIUPMSDTADGEGGEHTSNITASNOAMMEFYROWMTGHEEARPCHM
THDATOCOONNRECYTSATHIETXOPESOLFILTSSSICDERWAOLLCS

SHAPES AND ANGLES

1 Boxes are temporary walls that are built inside a proscenium stage at an angle to the original walls. Their purpose is to allow audiences sitting at the left or right of the stage to see the whole stage. On the diagram at the right, draw the most appropriate positions for the boxes in order to achieve this purpose.



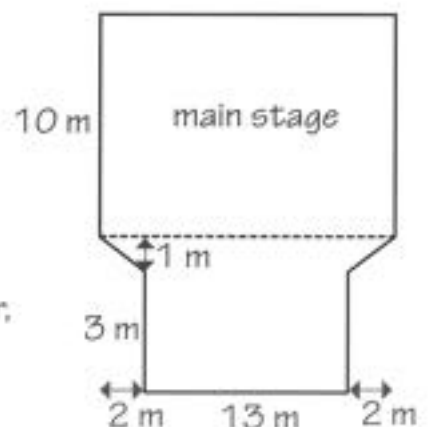
2 Below are 4 designs for an arena (or theatre in the round) drama performance. The stages are shaded grey. Describe each stage shape and the audience seating arrangement. Use the word 'concentric' in your descriptions. Then write one advantage and one disadvantage of each design.



3 This is a diagram of a thrust stage, which is a stage that extends into the audience from the main stage.

- What is the area of the main stage? _____
- How much area has been added with the thrust stage?

- What factors must be taken into account by a set designer, who constructs and positions props and backdrops, when planning a play on a thrust stage?



7 FOOD TECHNOLOGY

WORD LIST:

• ingredients	• survey	• percentage	• gram
• recipe	• label	• metabolism	• menu
• ratio	• table	• rate	• combination
• portion	• nutrition	• catering	• entree
• proportion	• market	• profit	• dessert
• amount	• research	• loss	• millilitre
			• cup
			• tablespoon
			• teaspoon
			• serving

Use the word list above to complete the following exercises.

TRIOS – Circle the correct term to complete each sentence.

- 1 In Australia, 2 tablespoons of oil is equal to **4 / 6 / 8** teaspoons.
- 2 There are approximately **15 / 20 / 25** millilitres to an Australian tablespoon.
- 3 A serving is the same as **a portion / a course / an amount**.
- 4 Four grams of fat in a 160 g tub of custard equals a percentage fat content of **1.25 / 2.5 / 5**.
- 5 Market research showed a 55% approval rating for a new product. This means that of the 300 people surveyed, **110 / 165 / 210** people approved the product.
- 6 A metric cup is equal to **100 / 250 / 500** mL.
- 7 A restaurant menu has a choice of 5 entrees, 8 main meals and 4 desserts. There are **17 / 105 / 160** different menu combinations at this restaurant.

SORT AND SEQUENCE – CATERING – For the information given, answer the questions below.

A school formal is organised. The meal will cost \$33 per head for a three-course meal and \$25 per head for a two-course meal. The venue costs \$1400 for the night for under 50 people and \$1600 for the night for 50 to 100 people. This amount includes DJ, music and security. It is expected that 80% of Year 12 students will attend as well as 16 teachers. There are 90 Year 12 students at the school. The teachers do not have to pay to attend.

- 1 What is the first decision that has to be made? Choose one. _____
- 2 What is the second number that needs to be determined? Calculate it. _____
- 3 Which two pieces of information are now not required? _____

- 4 What amount must now be determined? Calculate it. _____
- 5 How will the cost per student be calculated? What is it, correct to the nearest \$? _____

INTERPRETATIONS – Read the following and answer the questions below.

A jar of curry mix and a tin of lite coconut milk have the following nutrition labels:

	CURRY MIX	Servings: 3 Serving size: 135 g	LITE COCONUT MILK	Servings: 5 Serving size: 80 mL	
	Per serving	Per 100 g	Per serving lite milk	Per 100 mL standard milk	Per 100 mL lite
Fat total	14.2 g	10.5 g	4.8 g	17 g	6 g
– saturated*	9 g	6.6 g	4.0 g	12 g	5 g
Carbohydrates	8.1 g	6.0 g	0.6 g	1.2 g	0.7 g
– as sugars^	1.4 g	1.0 g	< 0.1 g	1 g	< 0.1 g
Sodium	567 mg	420 mg	8 mg	15 mg	10 mg

*Saturated fat raises blood cholesterol which chokes arteries and can lead to heart attacks and strokes. Total daily fat intake should be about 40 g, of which saturated fat should be less than 15 g.

^The recommended daily sugar intake is about 40 g.

The recipe on the curry mix jar recommends 300 g of meat of your choice and 200 g mixed vegetables, plus 1 cup of coconut milk for a milder curry.

- What percentage of the fat total in the curry mix is saturated? _____
 - What percentage of the fat total in the lite coconut milk is saturated? _____
 - Are these good figures? Why? _____
- What is the total weight of the curry mix in the jar? _____
- How many mL of coconut milk are in the tin? _____
- Are the sugar levels healthy? Why? _____
- if you choose lite coconut milk over standard:
 - What is the approximate ratio of fat reduction? _____
 - What is the approximate ratio of sodium reduction? _____
- What combination of numbers needs to be assessed when considering how much fat is in a product? _____

8 GEOGRAPHY

WORD LIST:

- map
- scale
- sentence
- ratio
- linear
- grid
- reference
- easting
- northing
- horizontal
- vertical
- topographic
- physical
- political
- thematic
- globe
- axis
- unit
- latitude
- longitude
- region
- representation
- projection
- temperature
- average
- annual
- precipitation
- hemisphere

Use the word list above to complete the following exercises.

MAPS

Scales – There are three ways to represent a scale on a map. A large scale map is more detailed than a small scale map.

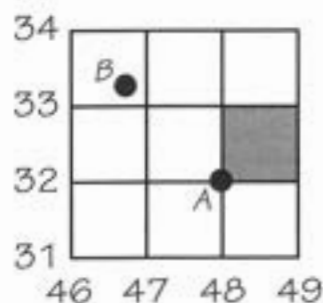
1 Complete the following table.

Map	Sentence scale	Ratio scale	Linear scale (km)
A	1 cm represents 50 km		
B		1:50 000	
C			

2 Which map from the table above would show the most detail – A, B or C? Why?

3 Which map from the table above would show the least detail – A, B or C? Why?

Grid references



4 Use the section of a topographic map shown at the left to answer these questions.

- a On which axis are the eastings? _____
- b On which axis are the northings? _____
- c What is the 4-figure grid reference for the shaded square?

- d What is the 6-figure grid reference for point A? _____
- e What is the 6-figure grid reference for point B? _____

- 5 Describe one major difference between using latitude and longitude and using other forms of coordinate positioning.

TRIOS – Circle the correct term to complete each sentence.

- 1 A topographic map would best display the **region** around the town of Tenterfield / **state** of Queensland / **Adelaide CBD**.
- 2 The minerals of Australia would appear on a **physical** / **political** / **thematic** map.
- 3 The countries of the world would appear on a **physical** / **political** / **thematic** map.
- 4 The mountain ranges of Asia would appear on a **physical** / **political** / **thematic** map.
- 5 The only accurate representation of the Earth is on a **map** / **globe** / **satellite image**.
- 6 A two-dimensional representation of the Earth is called a **map** / **projection** / **globe**.

CLIMATE GRAPHS – Refer to the climate graph to answer the following questions.

- 1 What type of graph represents rainfall?

- 2 What type of graph represents temperature?

- 3 What is the maximum annual temperature? In which months does it occur?

- 4 What is the minimum annual temperature? In which months does it occur?

- 5 In what unit is the temperature measured? _____

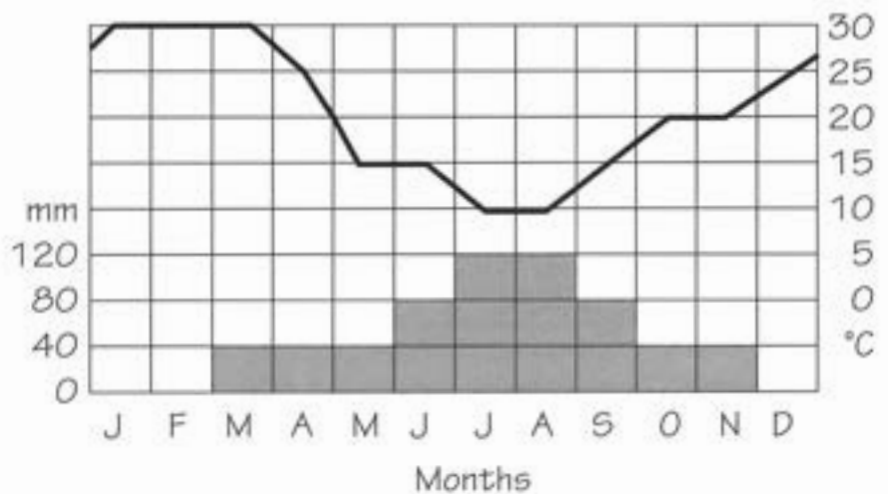
- 6 In what unit is precipitation measured? _____

- 7 What is the average annual temperature range? _____

- 8 What is the total average annual rainfall? _____

- 9 Is Station Y in the Northern Hemisphere or Southern Hemisphere? How do you know?

Climatic data for Station Y



9 MORE GEOGRAPHY

WORD LIST:

- knot
- debt
- currency
- fraction
- Beaufort
- nautical
- population
- moderate
- relationship
- navigators
- labour
- percentage
- wealthy
- tourism
- height
- imports
- ratio
- statistics
- agriculture
- effects
- exports
- exchange
- data
- industry
- average
- national
- rate
- immediate
- services

Use the word list above to complete the following exercises.

CASE STUDY

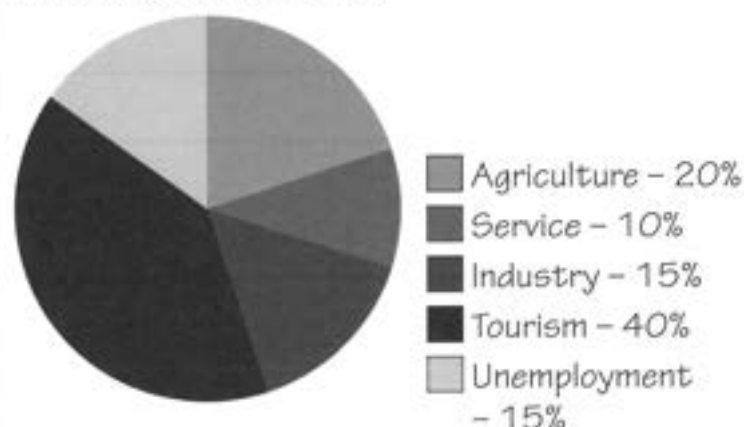
The Beaufort Scale was created centuries ago as a description of wind velocity, indicating what ships could expect in varying sailing winds. Its modified form is still used today by ocean navigators. Some of the scale is outlined in the table.

Beaufort force number	Description	Average wind speed in knots*	Land effects	Ocean effects	Wave heights at sea (m)
1	Light air	2	Smoke blows in wind direction	Sea ripples	0
4	Moderate breeze	13	Leaves move	Small waves	1
7	Near gale	31	Trees in motion	Foam from breaking waves	4
10	Storm	51	Trees uprooted	High waves, sea looks white	9
12	Hurricane	65	Serious damage	Little visibility, driving spray	15

*knots are nautical speeds, 1 knot = 1.852 km/h

Cabirin is a fictitious tropical island nation. The graph below shows how the island's labour force is employed. Consider also the statistics in the box below right.

Cabirin labour allocation



- Major exports: bauxite, sugar, bananas
- Major imports: machinery, fuel, food
- Exports: \$ A 1.8 billion
- Imports: \$ A 3 billion
- National Debt: \$ A 5 billion
- Population below poverty line: 20%
- Currency: Cabirin Bueno
- Exchange rate: \$ A 1 = 40 CB

- 1 Convert these wind speeds from knots to km/h.
- a Force 1: _____ c Force 7: _____ e Force 12: _____
- b Force 4: _____ d Force 10: _____
- 2 What do you think the description for a Force 8 wind would be? _____
- 3 Is there a linear relationship between the Beaufort Scale and wind speed? Explain your answer. _____
- 4 What words would be used to describe the wind and the ocean effects for a Beaufort Scale reading of Force 0? _____
- 5 Would it be safe to be on the water in a Force 5 wind? Explain. _____
- 6 What percentage of the Caborin labour force is employed? _____
- 7 To the nearest whole number, by how much do imports exceed exports in Caborin? _____
- 8 What do you think \$A stands for in this data? _____
- 9 How many Australian cents is 1 Caborin Bueno? _____
- 10 Do you think Caborin is a wealthy nation? Give at least three reasons for your answer. _____
- 11 Write the simple ratio of those employed in tourism to those employed in industry. _____
- 12 What fraction of the labour force works in agriculture? _____
- 13 Caborin is struck by a Force 12 wind. What does this mean? _____
- 14 For each industry, describe the immediate effects of the storm on the island and give reasons for your answers.
- a Tourism _____
- b Agriculture _____
- 15 Which industry will be least affected by the storm? Why? _____

10 HISTORY

WORD LIST:

- square
- volume
- rectangular
- label
- ancient
- construction
- height
- structure
- primary
- wonder
- destruction
- exist
- dimensions
- secondary
- base
- timeline
- pyramid
- key
- source
- approximate
- chronological

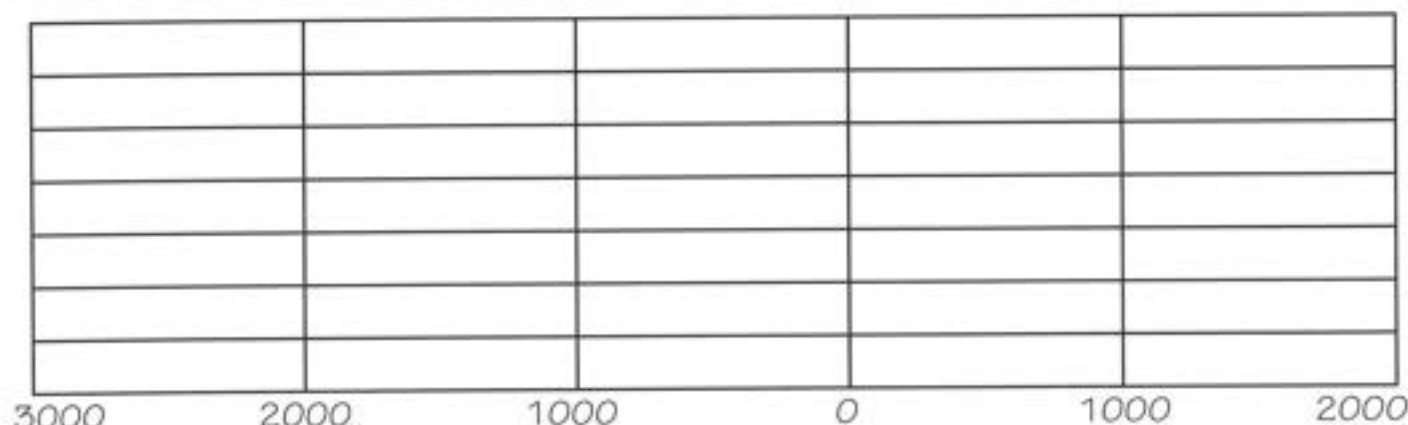
Use the word list above to complete the following exercises.

THE SEVEN ANCIENT WONDERS

Wonder	Constructed*	Destroyed*	Dimensions^
Great Pyramid at Giza	2580 BC	NA	230 m long x 230 m wide x 148 m high
Hanging Gardens of Babylon*	580 BC	80 BC	120 m long x 120 m wide x 24 m high
Statue of Zeus	460 BC	550 AD	Base 6.5 m square, 13 m high
Temple of Artemis	550 BC	356 BC	80 m long x 130 m wide x 20 m high
Mausoleum of Maussollos	350 BC	1490 AD	Base 40 m x 30 m, 40 m high
Colossus of Rhodes	286 BC	226 BC	34 m high
Lighthouse of Alexandria	280 BC	1390 AD	4 sections: base 27 m high, square 56 m high, octagonal 27 m high, circular 7 m high

*approximate mean dates, ^approximate, *built by Nebuchadnezzar as a gift for his wife Amytis.

TIMELINE – Using each row to represent a different ancient wonder, complete timelines from the year of construction to the year of destruction. Use different colours and a key. You also need to label each year axis appropriately.



Key:



TRIOS – Circle the correct term to complete each sentence.

- 1 The volume of the Great Pyramid is closest to **7 800 000 / 2 600 000 / 39 00 000** cm³.
- 2 The total height of the Lighthouse of Alexandria is **90 / 117 / 121** m.
- 3 The ancient wonder that existed for the shortest period of time was the **Statue of Zeus / Colossus of Rhodes / Temple of Artemis**.
- 4 The ancient wonder rectangular structure with the greatest volume was the **Temple of Artemis / Hanging Gardens of Babylon / Mausoleum of Maussollos**.
- 5 The existence of the Ancient Wonders is verifiable from **primary / secondary / primary and secondary** sources.
- 6 The Hanging Gardens of Babylon existed for **660 / 580 / 500** years.
- 7 The bronze Colossus of Rhodes was probably destroyed by **cannon / fire / earthquake**.

WHAT'S THE PURPOSE? – Write a sentence describing a possible purpose for each construction.

- 1 Great Pyramid: _____

- 2 Hanging Gardens: _____

- 3 Statue of Zeus: _____

- 4 Temple of Artemis: _____

- 5 Mausoleum of Maussollos: _____

- 6 Colossus: _____

- 7 Lighthouse of Alexandria: _____

11 MORE HISTORY

WORD LIST:

- | | | | | |
|----------------|--------------|---------------|-------------|----------|
| • statistics | • tonne | • population | • centuries | • mean |
| • Commonwealth | • average | • approximate | • arch | • height |
| • metre | • aggregate | • immigrant | • total | • length |
| | • percentage | | | |

Use the word list above to complete the following exercises.

NUMBERS – Use the Australian history statistics below to complete the cloze passage with the most appropriate word or number.

The Sydney Harbour Bridge

- | | |
|---|---|
| • Length of arch span: 503 m | • Height of top of arch: 134 m above mean sea level |
| • Width: 49 m | • Clearance for shipping: 49 m |
| • Total weight of arch: 39 000 t | • Height of pylons: 89 m above mean sea level |
| • Date opened: 15/11/1932 | |
| • Average number of cars crossing: 1932 – 10 900; 1943 – 8 600; 1991 – 181 900;
1992 – 138 400; 2001 – 159 600 | |

Cloze

The clearance of the Sydney Harbour Bridge for shipping, _____ metres, is the maximum _____ a ship can be _____ the water. The clearance is _____ to the width of the bridge. The length of the arch span is 503 _____ and the weight of the arch is _____.

The bridge was opened in _____ 1932 and the average, or _____, number of cars crossing it that year was _____. It dropped in 1943 because of the _____. By 1991, there were _____ vehicles _____. This dropped the _____ year due to the opening of the Sydney _____ Tunnel. These vehicle statistics have been rounded off to the nearest _____.

MORE NUMBERS

Immigration

In 2006, 4 960 000 of the Australian resident population was born overseas. This represents 24% of the total population. 1 150 000 of these Australians are from the United Kingdom. In 1901, 495 000 of the total 858 000 immigrants in Australia were from the UK.

- 1 What was the total Australian population in 2006? _____
- 2 What percentage of immigrants in 1901 was from the UK? _____

- 3 What percentage of immigrants in 2006 was from the UK? _____
- 4 Interpret what is demonstrated by this difference in percentage composition.

COMPRENDEZ – Read the passage and answer the questions below.

Grace Cossington-Smith was a modernist Australian artist whose paintings depicted the times as she lived them during the twentieth century.

Born on 22 April 1892, her first painting was 'Boy with a Cricket Bat' in 1908. 'The Sock Knitter' in 1915 supported Australians at war. 'The Prince' commemorated the visit to Sydney by the Prince of Wales in 1920. 'The Bridge in Curve' showed the approaching arches of the Sydney Harbour Bridge in 1930. 'Church Interior' in 1941 shows a church congregation where men are mostly absent. 'Interior with Veranda Doors' (1954) is a colourful painting of the newer Australian houses after the war. Her final painting was 'Still Life with White Cup and Saucer' in 1971.

Grace received an Order of Australia in 1983 and died on 10 December 1984.

- 1 How old was Cossington-Smith when she completed her first painting? _____
- 2 What is the meaning of the word *depicted*? _____
- 3 How old was Cossington-Smith when she completed her last painting? _____
- 4 Suggest why men were absent in 'Church Interior'. _____
- 5 What do you think 'The Sock Knitter' symbolises? _____
- 6 How old was Cossington-Smith when she died? _____
- 7 Complete this timeline for Cossington-Smith's birth, death and the paintings quoted in the passage.



12 METAL

WORD LIST:

- | | | | |
|-------------|----------------|-------------|---------------|
| • size | • drawing | • custom | • materials |
| • adjust | • design | • precise | • percentage |
| • measure | • modify | • calculate | • composition |
| • shape | • scale | • standard | • ratio |
| • malleable | • constituents | | |

Use the word list above to complete the following exercises.

COMPRENDEZ – Read the passage and answer the questions below.

Copper alloys

An alloy is generally a mixing of metals to form a more useful combination metal with properties often different to its constituents. Copper alloys have copper as their principal component. They are highly corrosion resistant.

Brass is an alloy of mainly copper and zinc. Gilding metal is 95% copper (Cu), 5% zinc (Zn), has a hardness* of 114 and is used for coins and ammunition. Muntz metal is 60% Cu, 40% Zn, has a hardness of 80 and is used in condenser tubes.

Bronzes are alloys of copper with tin, aluminium or silicon in particular. Phosphor bronze is 89% Cu, 10% tin (Sn) and 1% phosphorus (P), has a hardness of 241 and is used in springs and rivets.

Copper nickels and nickel silvers are alloys of mainly copper and nickel. Nickel silver is 76% Cu, 13% nickel (Ni), 9% lead (Pb), 2% tin (Sn), has a hardness of 55 and is used in ornaments and plumbing.

Copper is also alloyed with gold and silver to form precious metal alloys such as the blue-purple Shakudo.

**Hardness is a rating of a metal's resistance to deformation.*

- 1 What is the simple ratio of copper to zinc in Muntz metal? _____
- 2 If you have 400 g of Muntz metal, what is the net weight of copper? _____
- 3 Is an alloy harder when it has a higher hardness figure or a lower hardness figure? Justify your answer, using examples. _____

- 4 What do the chemical symbols Sn and Pb stand for? _____
- 5 Rewrite the first paragraph in simpler language. _____

- 6 Nickel silver has no silver in it. Why do you think it is called nickel silver? _____
- 7 Why would nickel silver be used in plumbing pipes? _____
- 8 If 2 tonnes of coins are made with gilding metal, how many kilograms of zinc would be used? _____
- 9 In which country do you think Shakudo originated? _____

IMPERIAL

Drill bits and other tools are made in both metric and imperial units. This is because countries such as the USA and Britain still use imperial units. It is important to be able to convert between them when reading imperial instructions and using metric tools. One inch equals approximately 25.4 mm.

- 1 Complete the table, rounding off your answers to the nearest mm.

- 2 Why are metric bits rounded off to the nearest mm? What problem can you see with rounding off the metric equivalents?
- _____
- _____

Drill bit imperial measurement	Equivalent drill bit metric measurement
$\frac{1}{2}$ "	
$\frac{3}{8}$ "	
$\frac{11}{16}$ "	
$\frac{7}{64}$ "	

WHAT'S THE DIFFERENCE? – What is the difference between:

- 1 a design and a scale drawing? _____
- 2 an estimate and a precise measure? _____
- 3 standard and custom? _____
- 4 modify and adjust? _____

13 MUSIC

WORD LIST:

• staff	• note	• clef	• instrument	• pitch
• horizontal	• parallel	• accompaniment	• ratio	• head
• phrase	• circle	• ledger	• frequency	• stem
• beats	• quaver	• chord	• rhythm	• flag
• melody	• semiquaver	• eighth	• octave	• hertz
• measure	• treble	• crotchet	• composition	• beam
	• bass	• repertoire	• meter	• lines

Use the word list above to complete the following exercises.

LABEL THAT / CLOZE – Use the word list to help complete the text below. Then use the words in bold to label figures 1 and 2.

1 The **staff** – where notes are written, made of 5 horizontal l _____ and 4 spaces.

2 The **meter** – the time of the music, represented by a f _____. The top number indicates how many b _____ in the bar (or measure) and the bottom number specifies the value of each note. In Figure 1, the notes in the m _____ will total $\frac{3}{4}$.

3 **Pitches** – named after the first 7 letters of the alphabet. In Figure 1, the pitch missing is _.

4 **Ledger lines** – additional lines above or below the staff p _____ to the staff.

5 A half n _____ – represented by a hollow circle (or head) and line (or stem).

6 An **eighth note** – represented by a full c _____ and stem and flag, also called a q _____.

7 The **clef** – a symbol at the beginning of the s _____ which determines the positioning of the pitches. The 2 main clefs are treble and b _____.

8 A quarter note is represented by a head and stem. It is also called a c _____. A sixteenth note is represented by a head, stem and 2 flags. It is also called a s _____ . Quavers and semi-quavers can be connected by a beam.

9 In Figure 2 above, there is a note missing from the second measure. Draw it in.

Figure 1

Figure 2

IMPERIAL

1 The strings section of the orchestra consists of four instruments. Complete the following table by converting the imperial units to metric (cm). The measures given are approximate. Use $1'' = 2.5$ cm and there are 12'' in 1 foot (1').

Instrument	Body size (imperial)	Body size (metric)	Full length (imperial)	Full length (metric)
Violin	14''		2'	
Viola	16''		28''	
Cello	30''		4'	
Double bass	x''	y cm	6'	

2 Assuming the ratio of body size to full size is consistent between cellos and double basses, what are the missing numbers x and y? _____

PUNCTUATION - Insert the correct punctuation, including quotation marks, and capital letters in the text below. Use [to indicate the start of a new paragraph.

my music teacher ms tune started the human ear perceives the frequency of sound waves as a pitch each musical note has a particular frequency does anyone know what these are measured in i raised my hand yes viola hertz thats correct she continued the average person can hear sounds between 20 Hz and 16 000 Hz ms tune walked to the board and wrote up the letter A the musical note A she explained has a frequency of 440 Hz she wrote that on the board each octave below will be half as many hertz again and each octave above will be twice as many hertz again how many hertz will the A one octave below this one be i volunteered to answer again ms tune pointed at me 220 Hz very good she complimented and can you tell me how many hertz two octaves above will be i quickly calculated it 1760 i replied thats right again well done this music stuff has a lot of maths in it i thought

BITS AND PIECES - Use these word parts to form 7 list words.

nlm me hm pe ti ord ph ac ra po rh re
yt ent si co ch rto dy on se lo mpa ire com

- | | |
|---------|---------|
| 1 _____ | 5 _____ |
| 2 _____ | 6 _____ |
| 3 _____ | 7 _____ |
| 4 _____ | |

14 PERSONAL DEVELOPMENT + PHYSICAL HEALTH

WORD LIST

- heart
- rate
- average
- maximum
- lifestyle
- communicable
- obesity
- formula
- weight
- height
- metres
- kilograms
- index
- fundamental
- movement
- locomotor
- non-locomotor
- manipulative
- combination

Use the word list above to complete the following exercises.

EXERCISING HEART RATES

The average maximum heart rate for an individual is found using the formula $M = (220 - \text{the person's age})$. It is recommended that in any exercise program, a person should aim for a heart rate between 50% and 80% of the maximum. An exercising heart rate should be within these target zones for health benefits to occur.

- 1 Complete the following table.

Age (years)	Maximum heart rate (beats per minute)	Lower target zone heart rate	Upper target zone heart rate
15			
30			
45			
60			

- 2 What do you notice about maximum heart rates as a person ages? _____

COMPRENDEZ – Read the following passage and answer the questions below.

Lifestyle diseases

Obesity is called a lifestyle disease because it is preventable. The Body Mass Index (BMI) is calculated by dividing your weight in kilograms by the square of your height in metres. A person with a BMI of more than 25 is considered overweight, and more than 30 as obese. About 54% of all Australians are overweight with a third of those obese. Obesity is now being referred to as an epidemic.

- 1 Complete the calculation of BMI as a mathematical formula using W for weight and H for height:

BMI = _____

- 2 Jessie is 1.5 metres tall and weighs 50 kilograms. Calculate her BMI. _____

- 3 Ronnie is 1.7 metres tall and weighs 82 kilograms. Calculate his BMI. _____
- 4 Compare Jessie's and Ronnie's BMI figures. _____

- 5 What percentage of the Australian population is obese? _____
- 6 What are some of the causes of obesity? _____

- 7 Circle any of the following conditions that obesity may increase the risk of suffering:
 - heart disease
 - hypertension
 - stroke
 - diabetes
 - ovarian cancer
 - prostate cancer
 - stress
 - low self-esteem
 - premature death
- 8 How is a lifestyle disease different from a communicable disease? _____

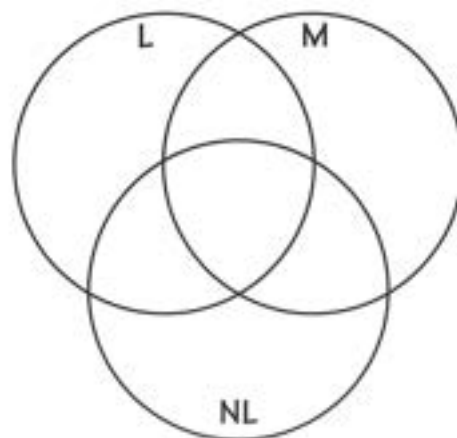
IT'S A VENN

Fundamental movement skills consist of three types of movement, outlined in the table. Activities can be based on one of these skills or a combination.

- 1 Using a full page in your workbook, copy the Venn diagram shown below right. Write each of the activities listed below in the most appropriate section of the Venn diagram.

Locomotor (L)	Manipulative (M)	Non-locomotor (NL)
run	throw	twist
hop	kick	turn
skip	control	balance
jump	bounce	pivot
gallop	catch	stretch

- kicking a field goal
 - a hurdle race
 - goal shooting in netball
 - swimming
 - a tennis serve
 - a pirouette
 - passing a ball
- playing handball
 - a spike in volleyball
 - archery
 - a broad jump
 - ten-pin bowling
 - throwing a javelin
 - taking a mark



15 PHOTOGRAPHY

WORD LIST:

- | | | | | |
|--------------|----------------|------------|---------------|--------------|
| • developing | • mixture | • increase | • convergence | • quality |
| • agent | • preservative | • exposure | • focus | • negative |
| • photograph | • litre | • aperture | • formula | • angle |
| | • factor | • focal | • replenish | • percentage |

Use the word list above to complete the following exercises.

COMPRENDEZ – Read the following passage and answer the questions below.

Developing a negative

A number of chemicals are used to produce a photograph. In its simplest form, the solution consists of:

- Developing Agents (DA) which convert a latent image to a visible image
- Alkali (A) which speeds up the reaction
- Preservative (P) which slows down oxidation of the DA.

To make 1 litre of Developing Formula (DF), 2 g of DA1, 100 g of P, 5 g of DA2, 2 g of A and up to 1000 cc of water are mixed.

To make 1 litre of Replenisher (R), which keeps the Developing Formula up to level as it is used, x g of DA1, 100 g of P, 7.5 g of DA2, 20 g of A and up to 1000 cc of water are mixed.

- 1 What does the abbreviation cc stand for? _____
- 2 By what factor does the amount of DA2 increase from the DF to R? _____
- 3 If the factor of increase of DA1 is the same, what is the missing value x in the Replenisher Formula? _____
- 4 What is the simple ratio of DA2 to P in DF? _____
- 5 What fraction of the Developing Formula chemicals is Preservative? _____
- 6 What is the percentage increase of the alkali from the Developing Formula to the Replenisher? _____
- 7 The mixing process begins with 750 cc of water heated to 50°C. Why do you think the water is heated at the start? _____
- 8 What is another way of describing the latent image? _____
- 9 Additives such as wetting agent and anti-sludging agent can be added to the mixture. What do you think these additives do? _____

10 In your own words, what does the Preservative do? _____

EXPOSURE

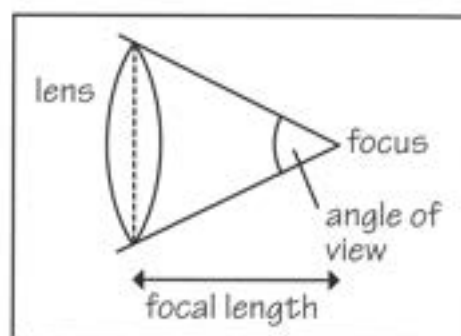
Aperture is the size of a hole which controls the amount of light reaching the film in a camera. It is expressed as an f-number. Standard f-numbers, in order of brightness, are f1.4, f2, f2.8, f4, f5.6, f8, f11, f16. Each f-stop allows half as much light as the previous one.

- 1 Which f-number allows the most light through? _____
- 2 Does f4 allow in twice as much or half as much light as f5.6? _____
- 3 Which f-stop allows in half as much light as f11? _____
- 4 Describe the mathematical pattern of the f-number sequence. _____

FOCAL LENGTH

The focal length in a camera is the distance from the lens to the point of convergence of light entering the lens, which is the focus.

Focal length	Use
21–35 mm	wide angle
35–70 mm	normal (angle of 50°)
70–135 mm	telephoto



- 1 Which lens is best for photographing a landscape? _____
- 2 Which focal length would be used for a portrait? _____
- 3 Which lens would have an angle of view of 20°? _____
- 4 Which focal length would have an angle of view of 60°? _____

BITS AND PIECES – Use these word parts to form 7 list words.

po gat pl ert rg sure tive ne op en vel ve rva
ure pr ap ex ish ence de con ing ese ive re

- 1 _____
- 2 _____
- 3 _____
- 4 _____
- 5 _____
- 6 _____
- 7 _____

WORD LIST:

• force	• potential	• Newtons	• freezes	• circuit
• mass	• difference	• molarity	• physics	• series
• acceleration	• current	• concentration	• chemistry	• energy
• distance	• resistance	• condenses	• solvent	• conductor
• speed	• ohms	• melts	• solute	• insulator
• time	• amperes	• boils	• solution	• magnetism
	• volts			

Use the word list above to complete the following exercises.

LABEL THAT

Three important physics formulas are:

$$F = ma$$

$$D = ST$$

$$V = IR$$

- 1 Using arrows, label each letter in the formulas above with the correct quantities and abbreviated units of measure.

Quantities: mass, speed, potential difference, force, time, current, acceleration, distance, resistance.

Abbreviated units of measure: V, m/s, m, N, kg, m/s², A, s, Ω.

- 2 If $m = 9.5$ and $a = 8$, what is the value of F ? _____
- 3 The distance a particle travels in 8 seconds is 440 metres. What is its average speed?

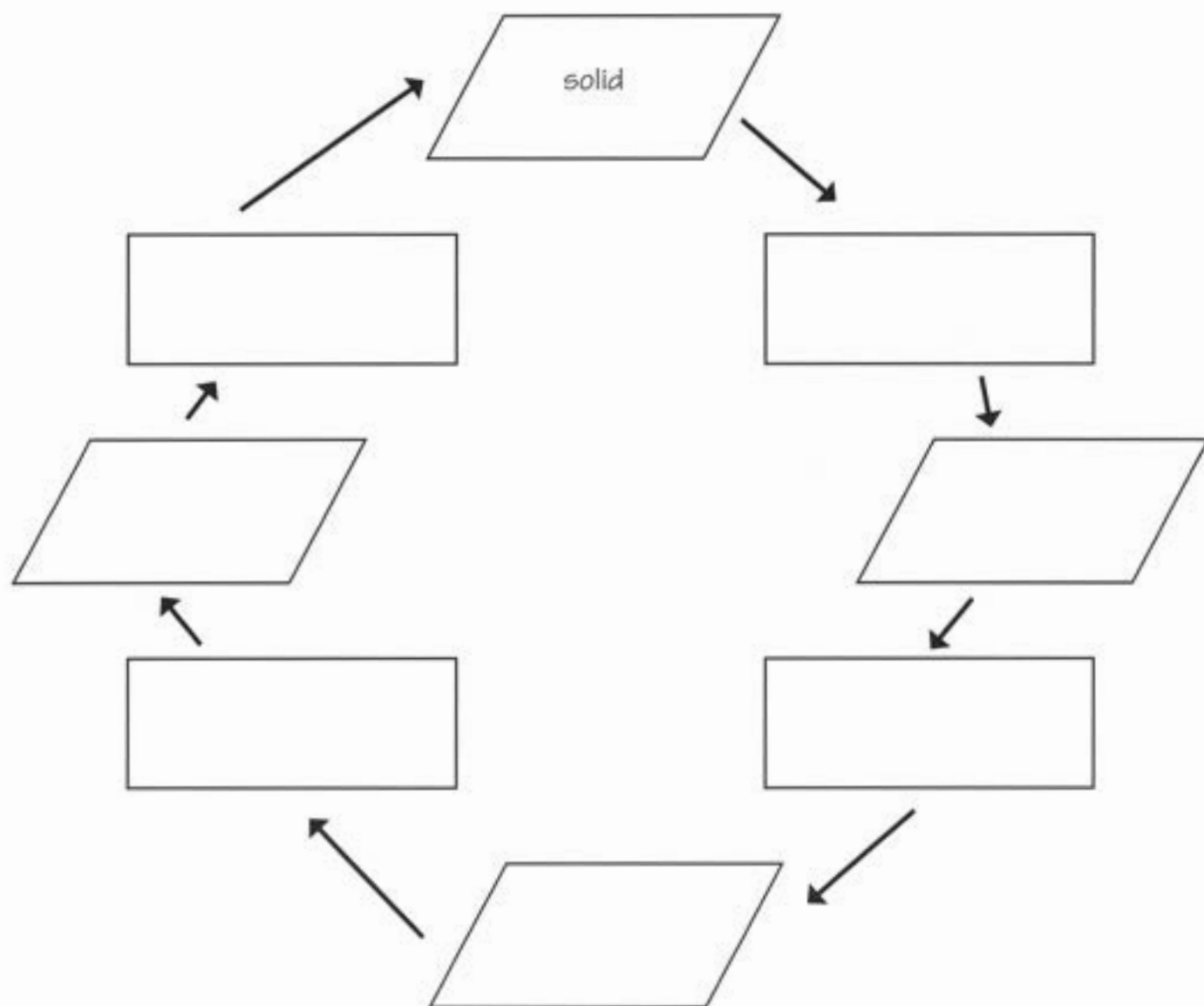
- 4 The potential difference across a circuit is 9V. If the resistance is 0.4Ω , what is the current through the circuit? _____

MOLARITY

In chemistry, molarity is a measure of the concentration of a solution, measured by how many moles of substance are diluted per litre of liquid. The mole unit is the total of the atomic weights of the elements (expressed in grams) that make up the substance being diluted.

- 1 Is 0.5 molar sulphuric acid more or less concentrated than 2 molar sulphuric acid? By what factor? _____
- 2 Hydrogen has an atomic weight of 1 and chlorine has an atomic weight of 35.5. How many grams in 1 mole of HCl? _____
- 3 Sodium has an atomic weight of 23, hydrogen 1 and oxygen 16. If 30 grams of NaOH is diluted in 1 litre of water, what is the molarity of the NaOH? _____

FLOW CHART – Complete the following change of state flow chart cycle. Using list words, write a state (e.g. solid) in each parallelogram and a process (e.g. boils) in each rectangle.



TRIOS – Circle the correct term to complete each sentence.

- 1 The average size of a male brain is **larger than** / **the same as** / **smaller than** the average female brain.
- 2 Convective and radiated describe forms of **heat** / **light** / **magnetism**.
- 3 The dissolved solid in a solution is called a **solvent** / **solute** / **suspension**.
- 4 If a second light is added to a series circuit, it **doubles the power usage** / **doubles the brightness** / **halves the brightness**.
- 5 Potential and kinetic are forms of **light** / **heat** / **energy**.
- 6 Most non-metals are **conductors** / **insulators** / **magnetic**.
- 7 An experiment tests a **hypothesis** / **an observation** / **a conclusion**.

17 MORE SCIENCE

WORD LIST:

- joules
- layer
- graph
- degrees
- prefix
- magnitude
- distance
- column
- exponential
- Richter
- scientific
- temperature
- approximation
- seismic
- scale
- notation
- thickness
- radius
- standard form
- energy

Use the word list above to complete the following exercises.

LARGE NUMBERS

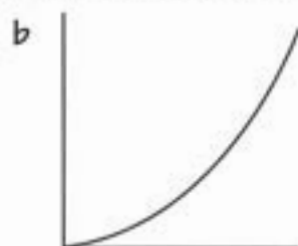
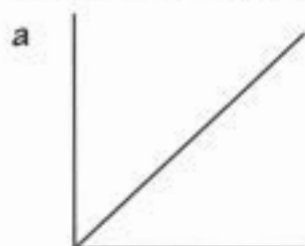
Large numbers are common in science. Prefixes are used to represent these numbers. Kilo means 1 000, or 10^3 . Mega means 1 000 000, or 10^6 . Other prefixes are listed in the table.

1 Write out the full numbers in the last column.

Prefix	Power of 10	Name	Number
Giga (G)	10^9	billion	
Tera (T)	10^{12}	trillion	
Peta (P)	10^{15}	quadrillion	
Exa (E)	10^{18}	quintillion	

THE RICHTER SCALE – The Richter Scale is an exponential scale that is used to measure the seismic energy release of an earthquake. 2, 4, 8, 16, 32 ... is an example of exponential growth.

1 Which of the following graphs best represents a Richter Scale?



A human body requires about 6 MJ of energy per day. Consider the following table.

Richter scale magnitude	Energy release in joules	Energy in joules in scientific notation
1	134 MJ	1.34×10^8
2	4.2 GJ	
4	4.2 TJ	
6	4.2 PJ	
9.3	134 EJ	

2 Complete the third column of the table.

- 3 For how many days can the energy release from a magnitude 1 earthquake supply the energy a human needs? _____
- 4 For how many days can the energy release from a magnitude 2 earthquake supply the energy a human needs? _____
- 5 A magnitude 6 earthquake releases the same energy as a 1 megaton bomb (1 million tonnes of TNT explosive). How many joules of energy in 1 tonne of TNT? _____
- 6 Rewrite the sentence below with correct spelling.
- The enegy prodused by the boxing Day Erthquake (and Sunami) 2004 (Rickter 9.3) coud have powered the USA four over 300 yaers.
- _____
- _____

LAYERS OF THE EARTH

- 1 Complete the table for the thickness of each layer.
- 2 Compare the temperature changes beneath the surface of the Earth and above it.
- _____
- _____
- _____
- 3 What is the radius of the Earth? _____
- 4 What is the approximate surface area of the Earth, written in scientific notation? (SA of sphere = $4\pi r^2$) Why is it an approximation?
- _____

Layer	Distance from Earth's surface (km)	Temperature (°C)	Thickness of each layer (km)
Exosphere	10 000	1000 to -270	
Thermosphere	690	Up to 20 000	
Mesosphere	85	-15 to -120	
Stratosphere	50	-60 to -15	
Troposphere	15*	-51 to 17	
Surface	0	-20 to 40	
Crust	20*	20 to 500	
Mantle	2 920	500 to 2000	
Outer Core	5 120	3000	
Inner core	6 380	4000	

*Average

- 5 If 70% of the Earth's surface is water, how many km² of land mass is there? Answer in standard form. _____

18 TEXTILES AND DESIGN

WORD LIST:

- grade
- percentage
- synthetic
- elastane
- canvas
- gram
- composition
- polyester
- satin
- ounce
- comparison
- constituents
- viscose
- cotton
- weight
- international
- natural
- nylon
- inches
- standard
- equivalent

Use the word list above to complete the following exercises.

IMPERIAL

Duck Canvas is used to make items such as hammocks, sand bags, boat covers, car-seat covers and some shower curtains.

Numbered Duck – if a standard cloth 36" by 22" weighs less than 19 ounces, the cloth is called Numbered Duck. The grade of Numbered Duck is calculated by the formula $G = 19 - W$, where W is the weight of the standard cloth in ounces.

Nought Duck – if a standard cloth 36" by 22" weighs more than 19 ounces, it is called Naught Duck. A $\frac{1}{10}$ Naught Duck weighs 19 ounces, $\frac{2}{10}$ Naught Duck weighs 20 ounces, and so on.

- 1 What is the area of a standard duck cloth in cm^2 ? Use $1" = 2.5 \text{ cm}$. _____
- 2 What is the grade of a Numbered Duck weighing 13 ounces? _____
- 3 What is the weight in ounces of a Grade 10 Numbered Duck? _____
- 4 If a standard cloth weighs 538.65 grams, how many grams in an ounce? _____
- 5 What is the weight of $\frac{5}{10}$ Naught Duck in ounces? _____
- 6 What is the weight of $\frac{5}{10}$ Naught Duck to the nearest gram? _____

WHAT'S IT MADE OF?

The table lists a variety of garments and their composition constituents.

Item number	Item of clothing	Percentage composition (from the label)
1	winter dress	66% polyester, 32% viscose, x% elastane
2	bike jacket	85% leather, 15% nylon
3	gym tights	54% polyester, 30% cotton, 16% elastane
4		90% satin, 10% cotton
5		100% nylon
6		100% wool
7		100% cotton

Polyester is usually a little heavier than cotton. Nylon is light and smooth. Elastane, or lycra, is very elastic material. Viscose, or rayon, breathes like cotton but is lighter. Satin is glossy.

A natural material comes from nature, while a synthetic material is manufactured by people.

1 Identify each material as natural (N) or synthetic (S).

a polyester__

d elastane__

f viscose__

b satin__

e leather__

g cotton__

c nylon__

2 In the table, what percentage of the winter dress is elastane?_____

3 The bike jacket weighs 1.6 kg. What weight is made up of leather? Answer in grams._____

4 There are 81 g of polyester in the gym tights. What do the gym tights weigh?_____

5 There are four items of clothing missing from the table. Write each clothing item in the most appropriate row, choosing from: t-shirt, swimming costumes, dress shirt, jumper.

SIZES

This table shows a comparison of international clothes sizes for ladies.

USA		4	6	8	10	12	14	16
UK	6	8	10	12	14	16	18	
Italy		38	40	42	44	46		
France	34	36	38	40	42	44		
Germany	32	34	36	38	40	42		
Australia	8	10	12	14	16	18		
Japan		7	9	11	13			
Singapore	4	6	8	10	12			

1 What is the equivalent Australian size for a Japanese 9?_____

2 What is the equivalent US size for an Australian 14?_____

3 What is the equivalent German size for a French 36?_____

4 If an Australian shops in Singapore, what is a basic rule to remember when sizing clothes?

5 If a UK tourist shops in Australia, what is a basic rule to remember when sizing clothes?

6 Is there any similar simple rule for an Italian shopping in Australia?_____

19 VISUAL ARTS

WORD LIST:

• brush	• technique	• distance	• artist	• irregular
• size	• tone	• falloff	• transformed	• polygon
• linear	• shading	• pencil	• tessellation	• create
• hatching	• parallel	• grade	• regular	• pyramid
	• perpendicular	• oblique		

Use the word list above to complete the following exercises.

SPELL MAD – Underline the incorrectly spelt words in the following passage. Then write the correct spelling above each one.

Brush sizes: Ther are twenty eight diferent sises of brush. Their are ten sizes abuve size twelv. Sising is liniar and a size twelve bruch requirs the hares from fourty eight squirrell tales.

Hatching: Hatching is a tecnique used in art to criate toan or shaiding affects by droring closly spased parrallel lines. if lines are drawed at a angel to one anuther, it is cauled cross-hatching. The clozer the lines are drew, the darcer the aria apears.

Distance falloff: Distanc fallof is a mefod of shading that maiks objekts futher away seam darkre. One wai of doing this is quadratic – a object twice as far away will recieve for times less lite.

Pencil grading: Pensils are graided from 9H, the hardest, then 8H, 7H, 6H, 5H, 4H, 3H, 2H, H, F, then HB, a medium grad, folowed by B, 2B, 3B and so on two 9B, the softest. H stands four hardnes, B meens blakness.

TRIOS – Circle the correct term to complete each sentence.

- 1 Linear sizing means each consecutive size is **equal / increasing / decreasing**.
- 2 A size 6 brush would require the hairs from **24 / 16 / 12** squirrels.
- 3 If cross-hatch lines are drawn at a 90° angle, the hatching is **parallel / perpendicular / oblique**.
- 4 Shading creates **depth / character / hatching** on a drawing.
- 5 Quadratic falloff means that **distance is proportional to the square of the light / distance is proportional to the inverse square of the light / light is proportional to the inverse square of the distance**.

6 Under this rule, if the distance between two objects is tripled, the further object receives **4 times less light / 6 times less light / 9 times less light.**

7 Soft pencils are **lighter / harder / blacker.**

8 The pencil 7 grades darker than 2H is **9H / 5B / 4B.**

COMPRENDEZ – Read the following passage and answer the questions below.

MC Escher

Maurits Cornelis Escher (1898–1972), was a famous Dutch artist who created unique works of art based on mathematics.

He specialised in transforming simple tessellations into complex and wondrous drawings. A tessellation is a repeating pattern that covers a page without gaps. He distorted regular and irregular polygons by inserting animals, buildings and other objects and flipped, glided and turned them to manufacture artworks that would otherwise have been dull except for his skilful touch.

Escher drew the Platonic solids (solids with faces of equal regular polygons) but placed some animal or object inside or around them to attract our attention. Or he would stellate them, meaning faces would be replaced with pyramids to make elaborate star shapes. He also constructed many 'impossible' drawings (illusions) that relied on the human brain for their effect.

Ants crawling along a Mobius strip is one example of Escher's best known work.

1 How old was Escher when he died? _____

2 What country did Escher come from? _____

3 Which 3 regular polygons can separately tessellate?

4 What are the mathematical terms for 'flip', 'glide' and 'turn'?

5 Write synonyms for the following words, as used in the passage.

a distorted _____

d wondrous _____

b manufacture _____

e elaborate _____

c complex _____

6 What is the Platonic solid based on:

a 4 equilateral triangles? _____

b 6 squares? _____

c 12 pentagons? _____

WORD LIST:

- wood
- size
- inch
- millimetre
- metric
- imperial
- width
- length
- thickness
- flooring
- veneer
- area
- planing
- square
- metre
- linear
- cut
- drill
- glue
- sand
- saw
- design
- chisel
- plane
- screw
- measure
- finish
- scale
- drawing
- variable

Use the word list above to complete the following exercises.

IMPERIAL – Wood sizes are often quoted in imperial units because the USA and UK use those units.

1 Complete the table by converting the measurements in inches to metric units. First round off the size in mm to the nearest whole unit. Then convert to cm and m using the mm answers. For this exercise, use 1 inch = 2.5 centimetres.

Wood type	Size in inches	Size in mm	Size in cm	Size in m
Cedar beam	$3\frac{1}{4}" \times 3\frac{1}{2}"$			
Spruce panel	$12" \times 36"$			
Oak sheet	$1\frac{1}{2}" \times 18" \times 18"$			
Pine plank	$1\frac{1}{2}" \times 4" \times 84"$			
Birch veneer	$1\frac{1}{8}" \times 48" \times 96"$			
Particle board	$5\frac{1}{8}" \times 60" \times 72"$			

LOST IN TRANSLATION

'I'd like to buy a 4 by 2 piece of wood please.' You would think that a 4 by 2 piece of wood has a width of 4" and a thickness of 2" and a variable length. However, the 4" by 2" is only the original size of the wood when rough sawn from the log. After drying and planing, its size is only $3\frac{1}{2}"$ by $1\frac{1}{2}"$.

1 a Explain how you would calculate the percentage volume that is lost between cutting and sale of a 4 by 2 piece of wood.

b Calculate the percentage. _____

c Why is its length irrelevant to answering this question? _____

WHAT'S THE ORDER? – You have been assigned the project of constructing a wooden tool box. Write the 15 steps for completing the project in a sensible order. For each step, write one of the terms below, and write an extra phrase to describe the step. For example, **costing** of materials.

Terms: mark, drill, chisel, plane, saw, design, measure, sand, glue, screw, scale drawing, purchase, size, finish, costing.

- | | |
|---------|----------|
| 1 _____ | 9 _____ |
| 2 _____ | 10 _____ |
| 3 _____ | 11 _____ |
| 4 _____ | 12 _____ |
| 5 _____ | 13 _____ |
| 6 _____ | 14 _____ |
| 7 _____ | 15 _____ |
| 8 _____ | |

HALF THE ALPHABET – Change the bold letter to form a different word. You must use a different letter for each one.

- | | |
|-------------------------|--------------------------|
| 1 w ood _____ | 8 s and _____ |
| 2 p laning _____ | 9 d esign _____ |
| 3 d rill _____ | 10 c oarse _____ |
| 4 g lue _____ | 11 s aw _____ |
| 5 s crew _____ | 12 p lane _____ |
| 6 s cale _____ | 13 d rawing _____ |
| 7 c ut _____ | |

A to Z – Write the list words in alphabetical order.

21 CURRICULUM CROSSWORD

ACROSS

- 2 In 2006, almost 5 million of the Australian resident population was born _____.
- 8 The average _____ heart rate for an individual is found using the formula $M = (220 - \text{the person's age})$.
- 9 In photography, what is the size of the hole which controls the amount of light reaching the film called?
- 10 Three synthetic materials are polyester, elastane and _____.
- 13 The focal length in a camera is the distance from the lens to the point of convergence of _____.
- 15 One important physics formula is:
 $\text{Force} = \text{mass} \times \text{_____}$.
- 16 Three natural materials are wool, silk and _____.
- 17 In music, pitches are named after the first 7 letters of the _____.
- 19 Wood sizes are often quoted in _____ units because the USA and UK use those units.
- 20 What is a mixture of metals called?
- 22 The Beaufort Scale is a description of wind _____.
- 24 A _____-_____ representation of the Earth is called a map.
- 25 In drama, this is a type of stage that extends into the audience from the main stage.
- 26 A type of lens that automatically changes the focal length and the angle of view from wide angle to telephoto.
- 29 A wage is an hourly rate of pay, and a _____ is an annual rate of pay.
- 30 _____ dance music is usually played between 120 and 200 beats per minute.
- 33 What is the scientific prefix that represents billion?
- 35 The recommended daily _____ intake is about 40 g.
- 36 There are 282 280 000 hectares of land used for _____ in Australia.
- 37 When using a potter's wheel to make an object, what sort of symmetry does the finished product have?
- 38 In drama, a temporary wall built inside a proscenium stage at an angle to the original walls is called a _____.

DOWN

- 1 1 inch equals approximately 2.54 _____.
- 3 The _____ Harbour Bridge was opened in the year 1932.
- 4 In musical notation, the staff is made up of 4 spaces and how many lines?
- 5 What is a repeating pattern that covers a page without gaps called?
- 6 MC Escher was a famous Dutch artist who created unique works of art based on _____.
- 7 Fundamental movement skills consist of three types of movement: locomotor, non-locomotor and _____.
- 11 Brass is an alloy made mainly of zinc and which other metal?
- 12 In chemistry, _____ is a measure of the concentration of a solution.
- 14 What type of scale is the Richter Scale?
- 18 The _____ of Alexandria is one of the Seven Ancient Wonders.
- 20 Nebuchadnezzar had the Hanging Gardens of Babylon constructed as a gift for his wife _____.
- 21 This type of fat raises blood cholesterol which chokes arteries and can lead to heart attacks and strokes.
- 23 A nautical unit of measurement, equal to 1.852 km/h.
- 25 Widescreen _____ and computer monitors have an aspect ratio of 16:9.
- 27 Financial advisers recommend that a family uses no more than 28% of the total gross annual income to pay a _____.
- 28 In the Czech Republic, a form of _____ was first made around 24 000 BC.
- 31 In the pH scale, is pH7 acidic, neutral or alkaline?
- 32 A type of dance that is in 4/4 time.
- 34 There are three ways to represent a scale on a map: sentence scale, ratio scale and _____ scale.



WORD LIST:

- acceleration
- agriculture
- alloy
- alphabet
- Amytis
- aperture
- box
- centimetres
- ceramic
- copper
- cotton
- electronic
- exponential
- five
- foxtrot
- giga
- imperial
- knot
- light
- lighthouse
- linear
- manipulative
- mathematics
- maximum
- molarity
- mortgage
- neutral
- overseas
- radial
- salary
- saturated
- sugar
- Sydney
- televisions
- tessellation
- thrust
- two-dimensional
- velocity
- viscose
- zoom

1 AGRICULTURE

The mathematics of feeding sheep: 1 a 7.8. b 8.4. c 929 g. 2 6.4, 2.7, 2370 g. 3 Food 1 as it has more than 3 times the MJ/kg. 4 50 kg sheep needs 6.8 MJ of ME, equalling 810 g of food $\times 1200 \times 30 = 29.16$ t.

Trios: 1 282 280 000. 2 grazing. 3 3/10. 4 45. 5 11 275 000. 6 154 544 000.

A to Z: acidity, agriculture, alkaline, animal, calculate, capacity, energy, hectares, kilogram, litres, megajoules, percentage, plant, storage, tonne, water, weight

2 CERAMICS

Clays: 1 37.5 min. 2 water. 3 0.96 kg. 4 8.2%. 5 9.6%.

The potter's wheel: 1 A circular shape along an axis of symmetry vertically through the centre of the top cross section. 2 Restricts the shapes of the pottery that can be made.

Which word?: 1 ceramics. 2 radial. 3 shrinkage. 4 materials. 5 pounds. 6 temperature. 7 composition. 8 texture.

A to Z: absorption, axis, ceramics, clay, composition, cone, materials, pounds, radial, shrinkage, symmetry, temperature, texture, thickness, vertical.

3 COMMERCE

Procedure: 1 divide, 320 000, 12.5. 2 divide, 12, 0.7. 3 calculate, 0.7, 280 000, 1960. 4 calculate $(7.8 \div 12)\%$ of 280 000 and subtract result from 1960, 140. 5 divide 1960 by $(93\ 000 \div 12)$, then multiply result by 100, 25.3. 6 monthly, 93 000, gross, multiplying, 11/3, 28, 2893. 7 subtract, 1960, 2893, 933.

What's the difference?: 1 Wage is hourly rate of pay, salary is annual rate of pay. 2 Temporary workers receive more work-related benefits such as sick pay and holiday pay but are paid less than casuals who don't receive benefits.

3 Commission is a percentage of sales, piecework is paid per unit. 4 Credit card pays with borrowed money, debit card pays with your own money. 5 Direct debit comes out of your account automatically, EFT you need to do manually. 6 Fixed rates stay the same for the term of the loan agreement, variable rates change depending on market conditions.

Every third letter: 1 comparative. 2 financial. 3 electronic. 4 temporary. 5 piecework. 6 mortgage. 7 variable. 8 employment. 9 adviser.

Half the alphabet: 1 g. 2 m. 3 a. 4 u. 5 r. 6 n. 7 s. 8 w. 9 l. 10 f. 11 t. 12 d.

4 DANCE

Staging a performance: 1 30 min 45 sec. 2 37 min. 3 8.47 pm. 4 hall hire, sound and light, catering and ushers, costumes and music. 5 \$5.

Musicality: 1 a 123 123. b 1234 1234 or 1234 5678.

2 a 4. b 128. c between each 32 bars. 3 Fast music and excited dancing. 4 a 64. b 2/4.

Direction of movement: 1 SE. 2 60° anticlockwise of 12. 3 left foot at 45° above right foot. 4 a reverse. b 360°. c 270°. d 450°.

5 DIGITAL MEDIA

Spell mad: automatically, focal, length, angle, angle, telephoto, An, zoom, maintains, resolution, meaning, number, clarity, be, magnifies, of, original, image, screen, pixels, sharpness, deteriorate, increasing, computer, in, picture, likely, pixels, overall, than, same.

Comprende: 1 1.78:1. 2 cropped. 3 letterbox. 4 pillar box. 5 window box. 6 Photo is slightly cropped as the ratio 4:3 is not the same as the ratio of the paper (3:2). 7 4:3 has area 1452 cm², widescreen has area 1293 cm²; 4:3 larger by 159 cm².

Image resolution: In table order – 0.21, 0.35, 2.07, 3.15, 70.00.

6 DRAMA

It's a numbers game: 1 Peter and Tom. 2 Swap Acts 2 and 7. 3 8.30 pm. 4 9.10 pm.

Every third letter: A proscenium stage gets its name from the arch that connects the tops of its side walls.

Shapes and angles: 1 Connect the top corners of the audience box to the side front corners of the stage and extend both lines to position the boxes. 2 Rectangular stage with concentric ellipses, square stage with concentric circles, rectangular stage with concentric rectangles, circular stage with concentric circles. 3 a 170m². b 54 m². c Must avoid having stage props in the audience view from the sides.

7 FOOD TECHNOLOGY

Trios: 1 8. 2 20. 3 a portion. 4 2.5. 5 165. 6 250. 7 160.

Sort and sequence: 1 2 or 3 course meal. 2 How many will attend, 80% of 90 + 16 = 88. 3 Venue cost for less than 50 and the cost of the number of courses not chosen. 4 total cost, \$3800 for 2 courses, \$4504 for 3 courses. 5 \$53 for 2 courses, \$63 for 3 courses.

Interpretations: 1 a 63%. b 83%. c No, percentage of saturated fat too high. 2 405 g. 3 400 mL. 4 Yes, low figures for both. 5 a 17:6. b 3:2. 6 Total per 100 units and the amount of saturated fats.

8 GEOGRAPHY

Maps: 1 A 1:5 000 000, linear scale in 50 km units.

B 1 cm represents 500 m, linear scale in 500 metre units, labelled in whole units every second gradation. C 1 cm represents 2 km, 1:200 000. 2 B as 1 cm = 500 m. 3 A as 1 cm = 50 km. 4 a horizontal. b vertical. c 4832. d 480 320. e 468 332. 5 Latitude is written first but it is the vertical axis unit where the horizontal axis is used in usual coordinates.

Trios: 1 region around the town of Tenterfield. 2 thematic. 3 political. 4 physical. 5 globe. 6 projection. **Climate graphs:** 1 column. 2 line. 3 30° in Jan Feb Mar. 4 10° in Jul Aug. 5 °C. 6 mm. 7 20°. 8 720 mm. 9 Southern as summer is start of year.

9 MORE GEOGRAPHY

Case study: 1 a 3.7. b 24.1. c 57.4. d 94.5. e 120.4. 2 gale. 3 no, wind speeds are not linear. 4 calm air, flat sea. 5 yes, not too rough yet. 6 85%. 7 \$1 200 000 000. 8 Australian Dollar. 9 2.5. 10 no, high poverty level, large national debt, high unemployment, poor exchange rate, imports food and machinery. 11 8:3. 12 1/5. 13 It was hit by a hurricane. 14 a Damage wrecks tourism industry. b Damage destroys crops. 15 Bauxite mining as the damage to mine sites likely to be minimal.

10 HISTORY

Trios: 1 2 600 000. 2 117. 3 Colossus of Rhodes. 4 Hanging Gardens of Babylon. 5 primary and secondary. 6 500. 7 earthquake.

What's the purpose?: 1 burial of pharaohs. 2 large structure with gardens in the desert as a gift. 3 large statue of king of the gods for worship. 4 to worship Artemus. 5 to bury Maussollos. 6 large structure to impress and intimidate. 7 protect ships from dangerous coast.

11 MORE HISTORY

Numbers: Cloze – 49, height, above, equal, metres, 39 000, tonnes, November, mean, 10 900, war, 181 900, crossing, next, Harbour, hundred.

More numbers: 1 20 666 667. 2 57.7. 3 23.2. 4 more immigrants from other countries now.

Comprende: 1 16. 2 represented. 3 79. 4 men at war. 5 knitting socks for soldiers. 6 92.

12 METAL

Comprende: 1 3:2. 2 240 g. 3 hardness figures increase with harder alloys. Copper nickels have lower hardness and are malleable, coins have a higher hardness and rivets even harder as they have to withstand great stresses. 4 tin, lead. 5 Example – If metals are mixed together to form another metal, it is called an alloy. Sometimes the alloy is different to the metals that are mixed to make it. The main part of a copper alloy is copper. Copper alloys don't corrode easily. 6 looks silver in colour. 7 easy to bend into shape, water won't corrode pipes. 8 100 kg. 9 Japan.

Imperial: 1 13, 10, 17, 3. 2 It would be silly to have bits with sizes such as 12.7 mm; the measures don't match very accurately.

What's the difference?: 1 Design is a sketch, a scale drawing is exact. 2 Estimate is approximate, precise is exact. 3 Standard is a regular size, custom is specially made. 4 Modify is on the whole design, adjust is on the finished product; modify is also on a larger scale where adjust is on a minor scale.

13 MUSIC

Label that/Cloze: 1 lines. 2 fraction, beats, measure. 3 g. 4 parallel. 5 note. 6 circle, quaver. 7 staff, bass. 8 crotchet, semi-quaver. 9 semi-quaver needs to be added in measure.

Imperial: violin – 35, 60. viola – 40, 70. cello – 75, 120. 2 double bass – 45, 112.5, 180.

Punctuation: My music teacher Ms Tune started. 'The human ear perceives the frequency of sound waves as a pitch. Each musical note has a particular frequency. Does anyone know what these are measured in?' I raised my hand. 'Yes Viola.'

'Hertz.'

'That's correct.' She continued. 'The average person can hear sounds between 20 Hz and 16 000 Hz.' Ms Tune walked to the board and wrote up the letter A. 'The musical note A,' she explained, 'has a frequency of 440 Hz.' She wrote that on the board. 'Each octave below will be half as many hertz again and each octave above will be twice as many hertz again. How many hertz will the A one octave below this one be?'

I volunteered to answer again. Ms Tune pointed at me. '220 Hz.'

'Very good,' she complimented, 'and can you tell me how many hertz two octaves above will be?'

I quickly calculated it. '1760,' I replied.

'That's right again. Well done.'

This music stuff has a lot of maths in it, I thought.

Bits and pieces: 1 accompaniment. 2 rhythm. 3 composition. 4 repertoire. 5 melody. 6 chord. 7 phrase.

14 PERSONAL DEVELOPMENT AND PHYSICAL HEALTH

Exercising heart rates: 1 15 – 205, 103, 164. 30 – 190, 95, 152. 45 – 175, 88, 140. 60 – 160, 80, 128. 2 decreases.

Comprende: 1 W + H². 2 22.2. 3 28.4. 4 Jessie's rating is good, Ronnie is overweight but not obese. 5 18. 6 lack of exercise, too much fat in diet, etc. 7 all of them. 8 Lifestyle diseases are self inflicted whereas communicable diseases are caught from other people.

It's a Venn: (suggested answers, discussions possible) in column order – M, M/L/NL, NL, M, L/M, L/NL, M, L/NL, M/NL, NL, M/L/NL, NL/M, L/M/NL, M/L.

15 PHOTOGRAPHY

Comprende: 1 cubic centimetre. 2 1.5. 3 3 g. 4 1:20. 5 1/10. 6 900%. 7 to help dissolve the solids. 8 invisible image. 9 keep the constituents in the mixture from hardening. 10 keeps the formula fresher for longer.

Exposure: 1 f1.4. 2 twice. 3 f16. 4 square roots of the powers of 2 – $\sqrt{2}$, $\sqrt{4}$, $\sqrt{8}$, $\sqrt{16}$, etc.

Focal length: 1 wide angle. 2 35–70. 3 telephoto. 4 21–35.

Bits and pieces: 1 exposure. 2 negative. 3 convergence. 4 replenish. 5 developing. 6 aperture. 7 preservative.

16 SCIENCE

Label that: 1 (from left) force (N), mass (kg), acceleration (m/s^2), distance (m), speed (m/s), time (s), potential difference (V), current (A), resistance (Ω). 2 76. 3 55 m/s. 4 22.5A

Molarity: 1 less by a factor of 4. 2 36.5 g. 3 0.75M.

Flow chart: clockwise – melts, liquid, boils, gas, condenses, liquid, freezes.

Trios: 1 larger than. 2 heat. 3 solute. 4 halves the brightness. 5 energy. 6 insulators. 7 a hypothesis.

17 MORE SCIENCE

Large numbers: 1 1 000 000 000, 1 000 000 000 000, 1 000 000 000 000 000, 1 000 000 000 000 000 000.

The Richter Scale: 1 b. 2 4.2×10^9 , 4.2×10^{12} , 4.2×10^{15} , 1.34×10^{20} . 3 221/3. 4 700. 5 4.2×10^9 joules. 6 energy, produced, Boxing, Earthquake, Tsunami, Richter, could, for, years.

Layers of the Earth: 1 9310, 605, 35, 35, 15, 0, 20, 2900, 2200, 1260. 2 Beneath increases in heat, above fluctuates between hot and cold. 3 6380 km. 4 $5.1 \times 10^8 \text{ km}^2$. 5 $1.5 \times 10^8 \text{ km}^2$.

18 TEXTILES AND DESIGN

Imperial: 1 4950. 2 6. 3 9. 4 28.35. 5 23. 6 652.

What's it made of?: 1 a S. b N. c S. d S. e N. f S. g N. 2 2%. 3 1360. 4 150 g. 5 dress shirt; swimming costumes; jumper; t-shirt.

Sizes: 1 12. 2 8. 3 34. 4 Singapore sizes 4 number sizes smaller than Australia. 5 Australian sizes 2 number sizes higher than UK. 6 Yes, add 28 to get Italian size.

19 VISUAL ARTS

Spell mad: There, eight, different, sizes, there, above, twelve, sizing, linear, brush, requires, hairs, forty, squirrel, tails, technique, create, tone, shading, effects, drawing, closely, spaced, parallel, If, drawn, an, angle, another, called, closer, drawn, darker, area, appears. Distance, falloff, method, makes, objects, further, seem, darker, way, doing, an, twice, receive, four, light. Pencils, graded, grade, followed, to, for, hardness, means, blackness.

Trios: 1 equal. 2 16. 3 perpendicular. 4 depth. 5 light is proportional to the inverse square of the distance. 6 9 times less light. 7 blacker. 8 4B.

Comprende: 1 74. 2 Holland. 3 square, equilateral triangle, regular hexagon. 4 reflection, translation, rotation. 5 a changed. b make. c complicated. d amazing. e intricate. 6 a square pyramid. b cube. c dodecahedron.

20 WOOD

Imperial: Cedar – 19 by 88, 1.9 by 8.8, 0.019 by 0.088. Spruce – 300 by 900, 30 by 90, 0.3 by 0.9.

Oak – $38 \times 450 \times 450$, $3.8 \times 45 \times 45$, $0.038 \times 0.45 \times 0.45$. Pine – $13 \times 100 \times 2100$, $1.3 \times 120 \times 240$, $0.003 \times 1.2 \times 2.4$. Birch – $3 \times 1200 \times 2400$, $0.3 \times 120 \times 240$, $0.003 \times 1.2 \times 2.4$. Particle – $16 \times 1500 \times 1800$, $1.6 \times 150 \times 180$, $0.016 \times 1.5 \times 1.8$.

Lost in translation: 1 a Subtract the smaller planed area from the original rough area, divide this answer by the original area and multiply by 100 to obtain the percentage loss. b 34.375%. c As the cross-sectional area of the wood is changing, it doesn't matter how long the wood is as the cross section is maintained through the full length.

What's the order? 1 design. 2 scale drawing. 3 costing. 4 purchase. 5 measure. 6 size. 7 mark. 8 saw. 9 chisel. 10 drill. 11 glue. 12 screw. 13 plane. 14 sand. 15 finish.

Half the alphabet: 1 f. 2 t. 3 g. 4 m. 5 h. 6 d. 7 o. 8 e. 9 r. 10 u. 11 l. 12 c. 13 p.

A to Z: area, chisel, coarse, cut, design, drawing, drill, finish, flooring, glue, imperial, inch, length, linear, measure, metre, metric, millimetre, plane, planing, sand, saw, scale, screw, size, square, thickness, veneer, width, wood

21 CURRICULUM CROSSWORD

Across: 2 overseas. 8 maximum. 9 aperture. 10 viscose. 13 light. 15 acceleration. 16 cotton. 17 alphabet. 19 imperial. 20 alloy. 22 velocity. 24 two-dimensional. 25 thrust. 26 zoom. 29 salary. 30 electronic. 33 giga. 35 sugar. 36 agriculture. 37 radial. 38 box.

Down: 1 centimetres. 3 Sydney. 4 five. 5 tessellation. 6 mathematics. 7 manipulative. 11 copper. 12 molarity. 14 exponential. 18 lighthouse. 20 Amytis. 21 saturated. 23 knot. 25 televisions. 27 mortgage. 28 ceramic. 31 neutral. 32 foxtrot. 34 linear.

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Commerce – rates, interest, income
Dance – time, fractions, compass points
Food technology – problem-solving, volume, capacity, cost
Geography – scale, grid reference, interpreting graphs

History – timelines, dimensions
Metal – percentage, converting units, Imperial system
Photography – capacity, length, number sequences
Science – using formulas, powers of 10, exponential scales
Textiles and design – mass, logical thinking

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